EN3160 - Assignment 02: Fitting and Alignment

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GitHub: A02_Fitting_and_Alignment_CV

In [52]:
```
from pathlib import Path

import cv2 as cv
import numpy as np
import matplotlib.pyplot as plt
from scipy.ndimage import gaussian_laplace, maximum_filter
from scipy.optimize import minimize
```

In [ ]:
```
image_dir = Path('a2images')
SAVE_FIGURES = False
output_dir = Path('results')

if SAVE_FIGURES:
    output_dir.mkdir(exist_ok=True)

plt.rcParams['figure.figsize'] = (9, 6)
plt.rcParams['font.size'] = 11
cv.setRNGSeed(0)
```

In [54]:
```
def read_image(path, flag=cv.IMREAD_COLOR):
    image = cv.imread(str(path), flag)
    if image is None:
        raise FileNotFoundError(f'Could not read {path}')
    return image


def show_image(ax, image, title):
    if image.ndim == 2:
        ax.imshow(image, cmap='gray', vmin=0, vmax=255)
    else:
        ax.imshow(cv.cvtColor(image, cv.COLOR_BGR2RGB))
    ax.set_title(title)
    ax.axis('off')


def save_figure(fig, name):
    if SAVE_FIGURES:
        fig.savefig(output_dir / name, dpi=200, bbox_inches='tight')
```

1. Sunflower detection using Laplacian of Gaussian

1.1 Read the sunflower image

In [ ]:
```
sunflowers = read_image(image_dir / 'the_berry_farms_sunflower_field.jpeg',cv.IM
gray = cv.cvtColor(sunflowers, cv.COLOR_BGR2GRAY).astype(np.float32) / 255.0
```

```
height, width = gray.shape
```

1.2 Build the LoG scale space

In [56]:
```
sigmas = np.geomspace(1.5, 20.0, 32)
responses = np.zeros((len(sigmas), height, width), dtype=np.float32)

for i, sigma in enumerate(sigmas):
    log_response = sigma**2 * gaussian_laplace(gray, sigma=sigma)
    responses[i] = np.maximum(log_response, 0)
```

1.3 Find maxima in space and scale

In [57]:
```
local_max = maximum_filter(responses, size=(3, 3, 3), mode='nearest')
field_start = int(0.50 * height)
response_threshold = 0.15 * responses[:, field_start:, :].max()
peak_mask = (responses == local_max) & (responses > response_threshold)
peak_mask[:, :field_start, :] = False

# A scale maximum needs neighbouring scales on both sides.
peak_mask[0] = False
peak_mask[-1] = False
peak_locations = np.argwhere(peak_mask)
```

1.4 Check for a dark centre surrounded by yellow petals

In [ ]:
```
hsv = cv.cvtColor(sunflowers, cv.COLOR_BGR2HSV)
yellow_mask = cv.inRange(hsv, (15, 100, 100), (40, 255, 255)) > 0
max_centre_intensity = 0.45
min_contrast = 0.10
min_yellow_fraction = 0.30
min_quadrant_fraction = 0.20


def has_flower_petals(x, y, radius):
    margin = int(np.ceil(2.0 * radius))
    if x - margin < 0 or x + margin >= width or y - margin < 0 or y + margin >=
        return False

    patch = gray[y-margin:y+margin+1, x-margin:x+margin+1]
    yellow = yellow_mask[y-margin:y+margin+1, x-margin:x+margin+1]
    yy, xx = np.mgrid[-margin:margin+1, -margin:margin+1]
    distances = np.hypot(xx, yy)
    centre = distances <= 0.55 * radius
    ring = (distances >= 1.15 * radius) & (distances <= 2.0 * radius)

    centre_mean = patch[centre].mean()
    if centre_mean > max_centre_intensity or patch[ring].mean() - centre_mean <
        return False
    if yellow[ring].mean() < min_yellow_fraction:
        return False

    quadrants = [(xx >= 0) & (yy >= 0), (xx < 0) & (yy >= 0),(xx < 0) & (yy < 0)
    supported = 0
    for quadrant in quadrants:
        part = ring & quadrant
        if part.any() and yellow[part].mean() >= min_quadrant_fraction:
```

```
            supported += 1
    return supported >= 3
```

1.4(b) Apply the petal check to the LoG candidates

In [59]:
```
candidates = []
for scale_index, y, x in peak_locations:
    radius = np.sqrt(2) * sigmas[scale_index]
    if not has_flower_petals(x, y, radius):
        continue
    score = responses[scale_index, y, x]
    candidates.append((float(x), float(y), float(radius),
                       float(sigmas[scale_index]), float(score)))

print(f'LoG peaks before filtering: {len(peak_locations)}')
print(f'Candidates passing the petal check: {len(candidates)}')
```

```
LoG peaks before filtering: 427
Candidates passing the petal check: 182
```

1.5 Remove duplicate detections

In [60]:
```
# Keep the stronger response when two circles describe the same neighbourhood.
candidates.sort(key=lambda item: item[4], reverse=True)
circles = []
for candidate in candidates:
    x, y, radius, sigma, score = candidate
    duplicate = False
    for old_x, old_y, old_radius, _, _ in circles:
        distance = np.hypot(x - old_x, y - old_y)
        if distance < 0.75 * (radius + old_radius):
            duplicate = True
            break
    if not duplicate:
        circles.append(candidate)

circles.sort(key=lambda item: item[2], reverse=True)
print(f'Sigma range: {sigmas[0]:.2f} to {sigmas[-1]:.2f} pixels')
print(f'Number of scales: {len(sigmas)}')
print(f'Positive LoG threshold: {response_threshold:.6f}')
print(f'Number of circles: {len(circles)}')
```

```
Sigma range: 1.50 to 20.00 pixels
Number of scales: 32
Positive LoG threshold: 0.060591
Number of circles: 140
```

1.6 Draw the detected circles

In [61]:
```
fig, axes = plt.subplots(1, 2, figsize=(14, 7))
show_image(axes[0], sunflowers, 'Reduced sunflower image')
show_image(axes[1], sunflowers, 'Dark flower centres (largest five numbered)')

for i, (x, y, radius, sigma, score) in enumerate(circles):
    colour = 'red' if i < 5 else 'lime'
    axes[1].add_patch(plt.Circle((x, y), radius, fill=False,
                                 color=colour, linewidth=1.2))
    if i < 5:
        axes[1].text(x, y, str(i + 1), color='white', fontsize=10,
```

```
                    ha='center', va='center', backgroundcolor='black')

fig.tight_layout()
save_figure(fig, 'q1_sunflower_circles.png')
plt.show()
```

1.7 Report the largest circle parameters

In [62]:
```
print('Largest dark-centre circles: coordinates are (x, y), measured from the to
print(f'{"No.":>4} {"x":>9} {"y":>9} {"radius":>9} {"sigma":>9}')
for i, (x, y, radius, sigma, score) in enumerate(circles[:5], start=1):
    print(f'{i:4d} {x:9.2f} {y:9.2f} {radius:9.2f} {sigma:9.2f}')

if not circles:
    print('No circles passed the filters. Inspect the image before adjusting the
```

```
Largest dark-centre circles: coordinates are (x, y), measured from the top left.
 No.         x         y    radius     sigma
   1    144.00    326.00     13.33      9.43
   2    278.00    232.00     11.28      7.98
   3    282.00    338.00     10.38      7.34
   4    152.00    254.00     10.38      7.34
   5    179.00    327.00      8.78      6.21
```

Discussion. Circles mark dark sunflower centres, not petals. Measurements use reduced-image pixels (multiply by four for original dimensions). Filtering reduces false detections but may miss small, shaded, tilted, hidden or edge flowers, so the count is incomplete. The previous 359-circle results were cleared. The revised detector remains untested; run it and inspect the five numbered circles before accepting measurements.

2. Line and circle fitting using RANSAC

Generate the noisy points

This uses the same distributions and parameters as Listing 1. A fixed random seed makes the point set repeatable. The true model labels are used only for plotting and later comparison, not to select the RANSAC inliers.

Generate the noisy circle points

Use the given centre, radius and radial noise.

In [63]:
```
rng = np.random.default_rng(0)
N = 100
half_n = N // 2

r_gt = 10.0
x0_gt, y0_gt = 2.0, 3.0
circle_noise = r_gt / 16
t = rng.uniform(0, 2 * np.pi, half_n)
n = circle_noise * rng.standard_normal(half_n)
x = x0_gt + (r_gt + n) * np.cos(t)
y = y0_gt + (r_gt + n) * np.sin(t)
X_circ = np.column_stack((x, y))
```

Generate the noisy line points

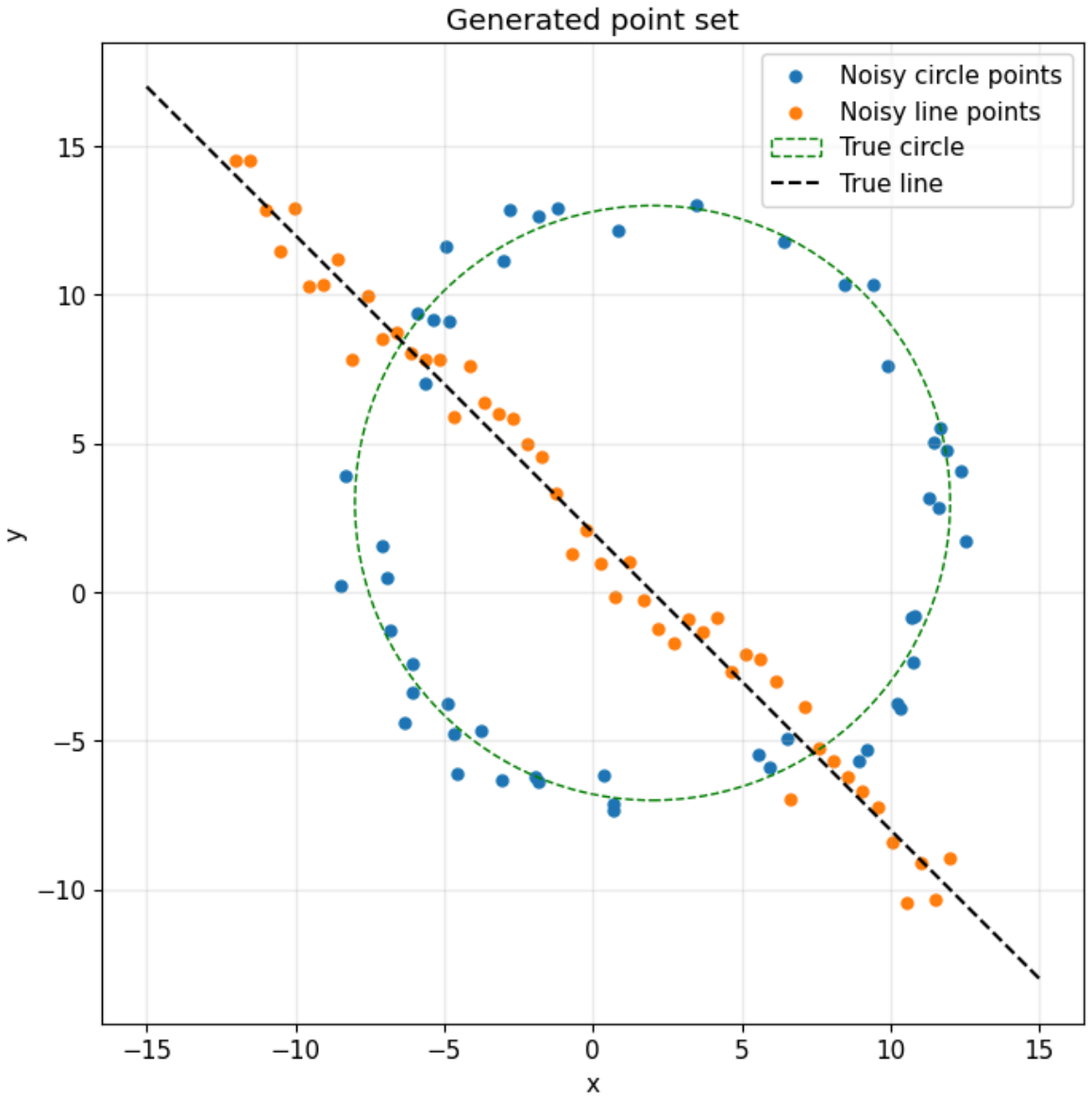
Add vertical noise to the line and combine both point sets into X.

In [64]:
```
m_gt, b_gt = -1.0, 2.0
line_noise = 1.0
x = np.linspace(-12, 12, half_n)
y = m_gt * x + b_gt + line_noise * rng.standard_normal(half_n)
X_line = np.column_stack((x, y))
X = np.vstack((X_circ, X_line))
```

Plot the generated data

Compare the noisy points with the two ground-truth models.

In [ ]:
```
fig, ax = plt.subplots(figsize=(8, 8))
ax.scatter(X_circ[:, 0], X_circ[:, 1], s=25, label='Noisy circle points')
ax.scatter(X_line[:, 0], X_line[:, 1], s=25, label='Noisy line points')
ax.add_patch(plt.Circle((x0_gt, y0_gt), r_gt, fill=False,color='green', linestyl
x_plot = np.linspace(-15, 15, 300)
ax.plot(x_plot, m_gt * x_plot + b_gt, 'k--', label='True line')
ax.set(xlabel='x', ylabel='y', title='Generated point set')
ax.set_aspect('equal')
ax.grid(alpha=0.25)
ax.legend()
plt.show()
```



2(a). RANSAC line

2(a).1 Calculate the number of RANSAC trials

In [ ]:
```
def ransac_trials(inlier_fraction, sample_size, confidence=0.999):
    return int(np.ceil(np.log(1 - confidence) /np.log(1 - inlier_fraction**sampl
```

2(a).2 Define a line and its error

Build a unit-normal line from two points, then measure perpendicular distances.

In [67]:
```
def line_from_two_points(points):
    p1, p2 = points
    direction = p2 - p1
    length = np.linalg.norm(direction)
    if length < 1e-10:
        return None

    normal = np.array([-direction[1], direction[0]]) / length
    d = normal @ p1
    if d < 0:
```

```
        normal = -normal
        d = -d
    return np.array([normal[0], normal[1], d])


def line_errors(points, model):
    a, b, d = model
    return np.abs(a * points[:, 0] + b * points[:, 1] - d)
```

2(a).3 Define the total least squares fit

Fit the final line to all points in a consensus set.

In [68]:
```
def fit_line_tls(points):
    centre = points.mean(axis=0)
    U = points - centre
    eigenvalues, eigenvectors = np.linalg.eigh(U.T @ U)
    normal = eigenvectors[:, np.argmin(eigenvalues)]
    normal = normal / np.linalg.norm(normal)
    d = normal @ centre
    if d < 0:
        normal = -normal
        d = -d
    return np.array([normal[0], normal[1], d])
```

2(a).4 Define the line RANSAC algorithm

Repeat sampling, error calculation and consensus selection.

In [69]:
```
def ransac_line(points, threshold, min_inliers, n_trials, seed=1):
    if len(points) < max(2, min_inliers):
        raise ValueError('Too few points for the required line consensus.')
    random = np.random.default_rng(seed)
    best_count = 0
    best_error = np.inf
    best_mask = None

    for _ in range(n_trials):
        indices = random.choice(len(points), size=2, replace=False)
        sample = points[indices]
        model = line_from_two_points(sample)
        if model is None:
            continue

        errors = line_errors(points, model)
        mask = errors < threshold
        count = int(mask.sum())
        if count < min_inliers:
            continue
        mean_error = np.mean(errors[mask]**2)

        if count > best_count or (count == best_count and mean_error < best_erro
            best_count = count
            best_error = mean_error
            best_mask = mask.copy()
            best_sample = sample.copy()
            best_sample_model = model.copy()
```

```
    if best_mask is None:
        raise ValueError('No line reached the minimum consensus. Check the param

    # Keep the winning consensus fixed while refitting, so the fit and sample ca
    best_fit = fit_line_tls(points[best_mask])
    return {'model': best_fit, 'sample_model': best_sample_model,
            'sample': best_sample, 'inliers': best_mask}
```

2(a).5 Fit the line

Choose the thresholds and call the RANSAC function.

In [70]:
```
line_threshold = 1.4
line_min_inliers = 35
line_n_trials = max(500, ransac_trials(0.35, 2))

line_result = ransac_line(X, line_threshold, line_min_inliers, line_n_trials)
line_model = line_result['model']
line_inliers = line_result['inliers']
a, b, d = line_model
```

2(a).6 Inspect the line parameters

Check the normal length, consensus size and perpendicular error.

In [71]:
```
print(f'Line trials: {line_n_trials}')
print(f'Best-fit line: {a:.5f} x + {b:.5f} y = {d:.5f}')
print(f'Normal length: {np.hypot(a, b):.6f}')
print(f'Winning line consensus: {line_inliers.sum()} points')
print(f'Perpendicular RMSE on that consensus: '
      f'{np.sqrt(np.mean(line_errors(X[line_inliers], line_model)**2)):.4f}')
```

```
Line trials: 500
Best-fit line: 0.71710 x + 0.69697 y = 1.46671
Normal length: 1.000000
Winning line consensus: 57 points
Perpendicular RMSE on that consensus: 0.7129
```

2(b). RANSAC circle after removing the line consensus

2(b).1 Define a circle through three points

Solve for the circumcentre and reject nearly collinear samples.

In [72]:
```
def circle_from_three_points(points):
    p1, p2, p3 = points
    A = 2 * np.vstack((p2 - p1, p3 - p1))
    rhs = np.array([p2 @ p2 - p1 @ p1, p3 @ p3 - p1 @ p1])

    # Nearly collinear points give an unstable circumcircle.
    if np.linalg.cond(A) > 1e6:
        return None
    centre = np.linalg.solve(A, rhs)
    radius = np.linalg.norm(p1 - centre)
    model = np.array([centre[0], centre[1], radius])
    if not np.all(np.isfinite(model)) or radius <= 0:
```

```
        return None
    return model
```

2(b).2 Define radial error and the fitting objective

The optimizer will minimize the sum of squared radial errors.

In [73]:
```
def circle_errors(points, model):
    xc, yc, radius = model
    distances = np.hypot(points[:, 0] - xc, points[:, 1] - yc)
    return np.abs(distances - radius)


def circle_objective(model, points):
    errors = circle_errors(points, model)
    return np.sum(errors**2)
```

2(b).3 Define the circle RANSAC algorithm

Select the strongest consensus, then refine its centre and radius.

In [ ]:
```
def ransac_circle(points, threshold, min_inliers, n_trials, seed=2):
    if len(points) < max(3, min_inliers):
        raise ValueError('Too few points for the required circle consensus.')
    random = np.random.default_rng(seed)
    best_count = 0
    best_error = np.inf
    best_mask = None

    for _ in range(n_trials):
        indices = random.choice(len(points), size=3, replace=False)
        sample = points[indices]
        model = circle_from_three_points(sample)
        if model is None:
            continue

        errors = circle_errors(points, model)
        mask = errors < threshold
        count = int(mask.sum())
        if count < min_inliers:
            continue
        mean_error = np.mean(errors[mask]**2)

        if count > best_count or (count == best_count and mean_error < best_erro
            best_count = count
            best_error = mean_error
            best_mask = mask.copy()
            best_sample = sample.copy()
            best_sample_model = model.copy()

    if best_mask is None:
        raise ValueError('No circle reached the minimum consensus. Check the par

    result = minimize(circle_objective, best_sample_model,
                      args=(points[best_mask],), method='L-BFGS-B',
                      bounds=[(None, None), (None, None), (1e-6, None)])
    if not result.success:
        raise RuntimeError(f'Circle refinement did not converge: {result.message
```

```
    return {'model': result.x, 'sample_model': best_sample_model,'sample': best_
```

2(b).4 Remove the line consensus and fit a circle

Only the remaining points are passed to circle RANSAC.

In [ ]:
```
remaining_indices = np.flatnonzero(~line_inliers)
remaining_points = X[remaining_indices]

circle_threshold = 1.25
circle_min_inliers = 25
circle_n_trials = max(1000, ransac_trials(0.35, 3))
circle_result = ransac_circle(remaining_points, circle_threshold,circle_min_inli
circle_model = circle_result['model']
```

2(b).5 Identify circle inliers and report the fit

Convert the remaining-point mask back to the original indexing before reporting the result.

In [ ]:
```
# Convert the circle mask back to the indexing of the complete point set.
circle_inliers = np.zeros(len(X), dtype=bool)
circle_inliers[remaining_indices[circle_result['inliers']]] = True
xc, yc, radius = circle_model

print(f'Points remaining after removing the line: {len(remaining_points)}')
print(f'Circle trials: {circle_n_trials}')
print(f'Best-fit centre: ({xc:.4f}, {yc:.4f})')
print(f'Best-fit radius: {radius:.4f}')
print(f'Winning circle consensus: {circle_inliers.sum()} points')
print(f'Radial RMSE on that consensus: 'f'{np.sqrt(np.mean(circle_errors(X[circl
print(f'Centre error from ground truth: {np.hypot(xc-x0_gt, yc-y0_gt):.4f}')
print(f'Radius error from ground truth: {abs(radius-r_gt):.4f}')
```

```
Points remaining after removing the line: 43
Circle trials: 1000
Best-fit centre: (1.8783, 2.9555)
Best-fit radius: 10.1238
Winning circle consensus: 43 points
Radial RMSE on that consensus: 0.5848
Centre error from ground truth: 0.1295
Radius error from ground truth: 0.1238
```

2(c). Combined plot

2(c).1 Define a line plotting helper

Handles a vertical line.

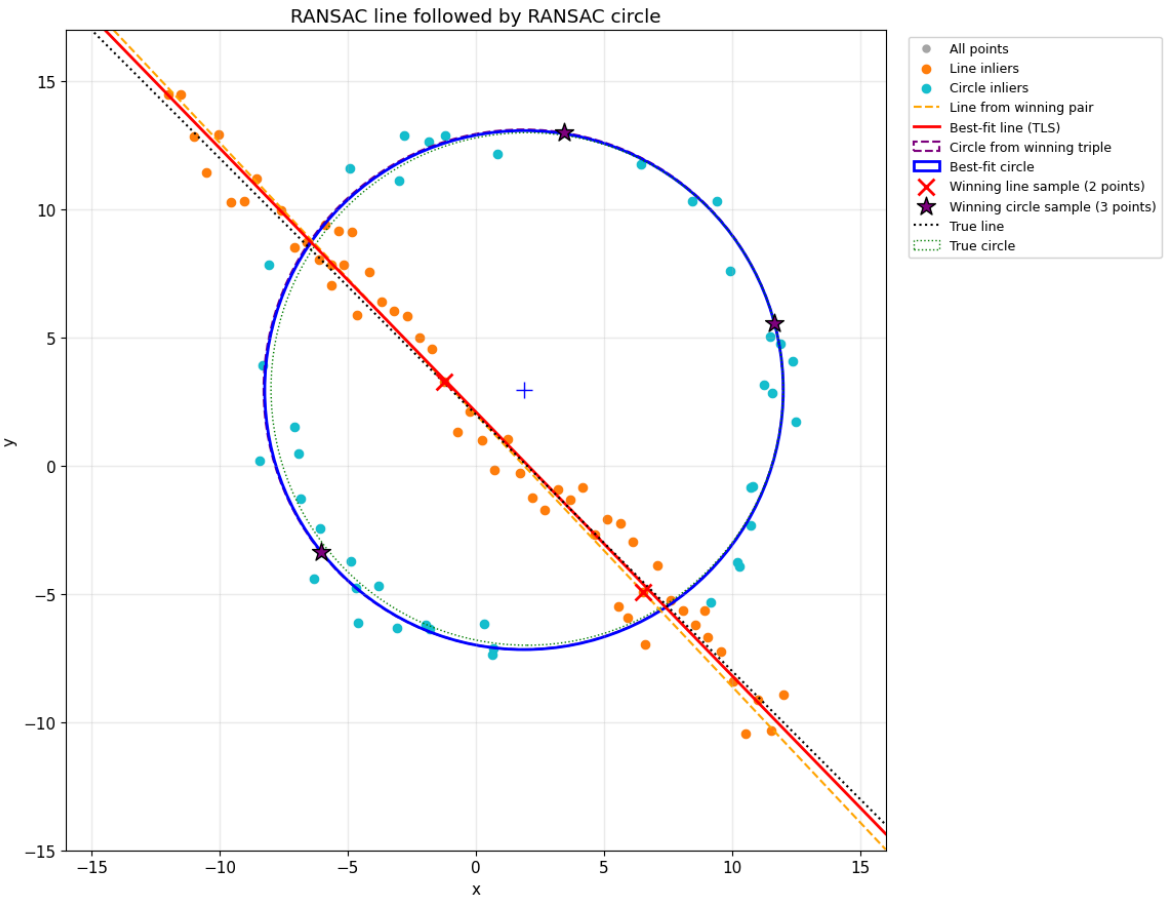
In [77]:
```
def draw_line(ax, model, **kwargs):
    a, b, d = model
    if abs(b) > 1e-10:
        xx = np.array([-16.0, 16.0])
        ax.plot(xx, (d - a * xx) / b, **kwargs)
```

```
    else:
        ax.axvline(d / a, **kwargs)
```

2(c).2 Compare the samples and fitted models

In [ ]:

```
fig, ax = plt.subplots(figsize=(11, 9))
ax.scatter(X[:, 0], X[:, 1], c='0.65', s=25, label='All points')
ax.scatter(X[line_inliers, 0], X[line_inliers, 1], c='tab:orange',s=35, label='L
ax.scatter(X[circle_inliers, 0], X[circle_inliers, 1], c='tab:cyan',s=35, label=

draw_line(ax, line_result['sample_model'], color='orange', linestyle='--',linewi
draw_line(ax, line_model, color='red', linewidth=2, label='Best-fit line (TLS)')

sample_xc, sample_yc, sample_r = circle_result['sample_model']
ax.add_patch(plt.Circle((sample_xc, sample_yc), sample_r, fill=False,color='purp
                        label='Circle from winning triple'))
ax.add_patch(plt.Circle((xc, yc), radius, fill=False, color='blue',linewidth=2,

line_sample = line_result['sample']
circle_sample = circle_result['sample']
ax.scatter(line_sample[:, 0], line_sample[:, 1], marker='x', c='red',s=130, line
ax.scatter(circle_sample[:, 0], circle_sample[:, 1], marker='*', c='purple',s=18

true_line = np.array([1, 1, 2]) / np.sqrt(2)
draw_line(ax, true_line, color='black', linestyle=':', label='True line')
ax.add_patch(plt.Circle((x0_gt, y0_gt), r_gt, fill=False, color='green',linestyl
ax.plot(xc, yc, '+', color='blue', markersize=12)
ax.set(xlim=(-16, 16), ylim=(-15, 17), xlabel='x', ylabel='y',title='RANSAC line
ax.set_aspect('equal')
ax.grid(alpha=0.25)
ax.legend(loc='upper left', bbox_to_anchor=(1.02, 1), fontsize=9)
fig.tight_layout()
save_figure(fig, 'q2_line_and_circle.png')
plt.show()
```



2(d). What happens if the circle is fitted first?

2(d).1 Fit the circle first and inspect its consensus

In [79]:
```
circle_first = ransac_circle(X, circle_threshold, 35, circle_n_trials, seed=12)
first_mask = circle_first['inliers']
first_xc, first_yc, first_radius = circle_first['model']

print('Circle fitted before removing the line:')
print(f'Centre = ({first_xc:.4f}, {first_yc:.4f}), radius = {first_radius:.4f}')
print(f'Circle-generated points in its consensus: {first_mask[:half_n].sum()} /
print(f'Line-generated points in its consensus: {first_mask[half_n:].sum()} / {h
print(f'Points left for a line fit: {(~first_mask).sum()}')
```

```
Circle fitted before removing the line:
Centre = (1.9461, 2.8866), radius = 10.1063
Circle-generated points in its consensus: 49 / 50
Line-generated points in its consensus: 8 / 50
Points left for a line fit: 43
```

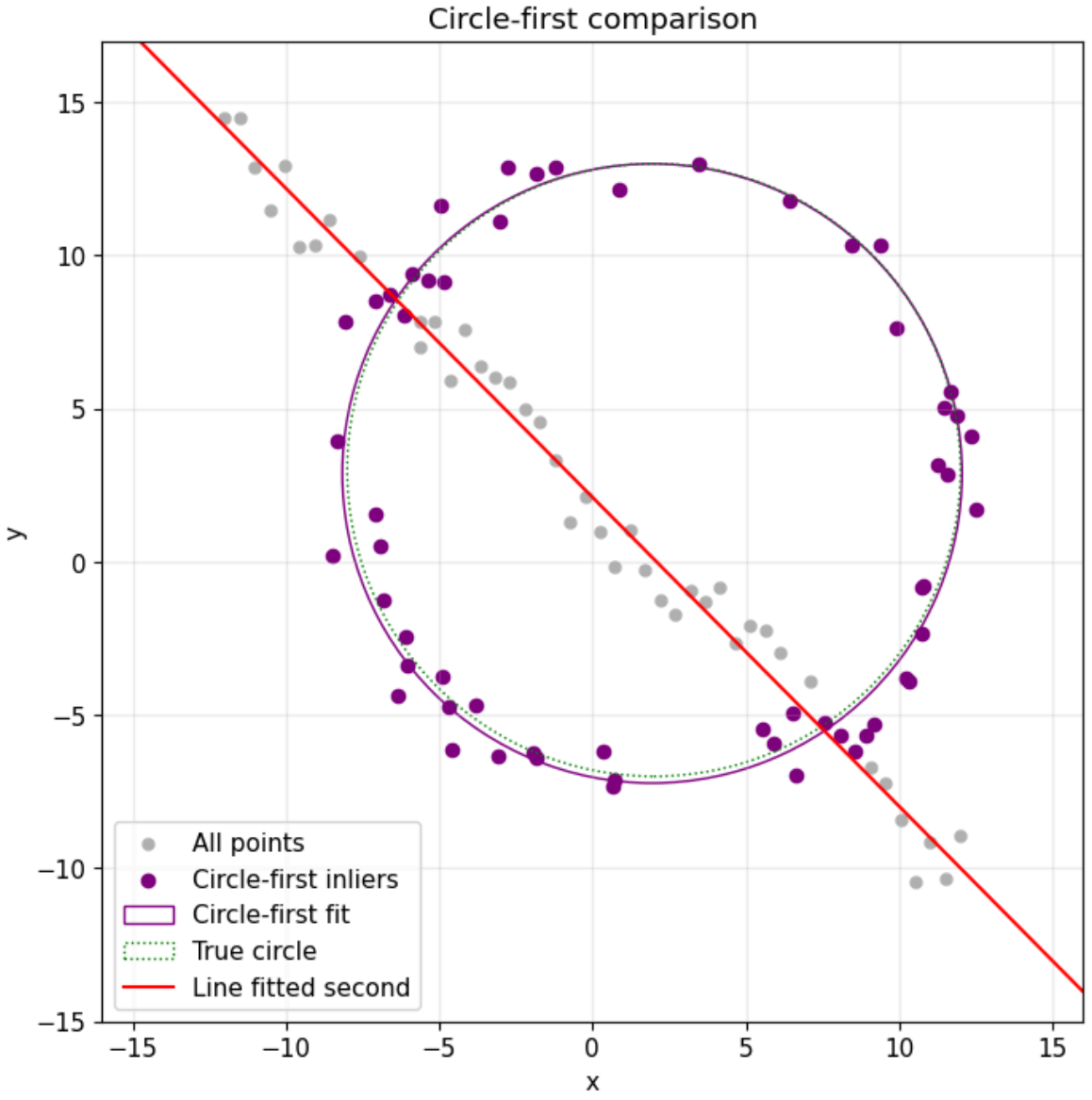
2(d).2 Fit the remaining line and plot the comparison

Check whether enough line points remain after removing the circle consensus.

In [ ]:
```
fig, ax = plt.subplots(figsize=(8, 8))
ax.scatter(X[:, 0], X[:, 1], c='0.7', s=25, label='All points')
ax.scatter(X[first_mask, 0], X[first_mask, 1], c='purple', s=35,label='Circle-fi
ax.add_patch(plt.Circle((first_xc, first_yc), first_radius, fill=False,color='pu
ax.add_patch(plt.Circle((x0_gt, y0_gt), r_gt, fill=False, color='green',linestyl
```

```
try:
    line_second = ransac_line(X[~first_mask], line_threshold,line_min_inliers, l
    draw_line(ax, line_second['model'], color='red', label='Line fitted second')
    print(f'Line consensus after fitting the circle first: {line_second["inliers
except ValueError as error:
    print(f'The remaining points did not support the required line consensus: {e

ax.set(xlim=(-16, 16), ylim=(-15, 17), xlabel='x', ylabel='y',
       title='Circle-first comparison')
ax.set_aspect('equal')
ax.grid(alpha=0.25)
ax.legend()
save_figure(fig, 'q2_circle_first.png')
plt.show()
```

```
Line consensus after fitting the circle first: 43
```



3. Homography and image blending

3.1 Define the four-point selection function

In [ ]:
```
def select_four_points(image, title, suggested_points=None):
    # Resize only the selection window. Convert clicks back to original coordina
```

```
    scale = min(1.0, 1000 / image.shape[1], 700 / image.shape[0])
    preview = cv.resize(image, None, fx=scale, fy=scale)
    scale_x = preview.shape[1] / image.shape[1]
    scale_y = preview.shape[0] / image.shape[0]
    if suggested_points is not None:
        guide = np.round(suggested_points * [scale_x, scale_y]).astype(np.int32)
        cv.polylines(preview, [guide], True, (0, 255, 0), 2)
        for i, (gx, gy) in enumerate(guide):
            cv.putText(preview, str(i + 1), (gx + 5, gy - 5),cv.FONT_HERSHEY_SIM
    points = []
    window = title + ' - TL, TR, BR, BL | Enter: accept | R: reset | Esc: cancel

    def mouse_callback(event, x, y, flags, param):
        if event == cv.EVENT_LBUTTONDOWN and len(points) < 4:
            points.append((x, y))

    cv.namedWindow(window, cv.WINDOW_AUTOSIZE)
    cv.setMouseCallback(window, mouse_callback)
    try:
        while True:
            canvas = preview.copy()
            for i, (x, y) in enumerate(points):
                cv.circle(canvas, (x, y), 5, (0, 0, 255), -1)
                cv.putText(canvas, str(i + 1), (x + 7, y - 7),cv.FONT_HERSHEY_SI
            if len(points) > 1:
                cv.polylines(canvas, [np.array(points, dtype=np.int32)],len(poin
            cv.imshow(window, canvas)
            key = cv.waitKey(20) & 0xFF
            if key in (10, 13) and len(points) == 4:
                break
            if key in (ord('r'), ord('R')):
                points.clear()
            if key == 27 or cv.getWindowProperty(window, cv.WND_PROP_VISIBLE) <
                raise RuntimeError('Point selection cancelled.')
    finally:
        cv.destroyAllWindows()

    return np.float32(points) / np.float32([scale_x, scale_y])
```

3.2 Define homography warping and blending

Map the source corners to the selected plane and blend using a separate mask.

In [ ]:

```
def place_overlay(background, overlay, destination, alpha=0.8):
    destination = np.float32(destination)
    if destination.shape != (4, 2):
        raise ValueError('Four destination points are needed.')
    contour = destination.reshape(-1, 1, 2)
    if not cv.isContourConvex(contour) or abs(cv.contourArea(contour)) < 10:
        raise ValueError('Select a convex quadrilateral in TL, TR, BR, BL order.

    source_height, source_width = overlay.shape[:2]
    source = np.float32([[0, 0], [source_width - 1, 0],[source_width - 1, source
    H = cv.getPerspectiveTransform(source, destination)
    height, width = background.shape[:2]

    warped = cv.warpPerspective(overlay, H, (width, height))
    source_mask = np.full((source_height, source_width), 255, dtype=np.uint8)
    mask = cv.warpPerspective(source_mask, H, (width, height),flags=cv.INTER_NEA
```

```
    # Warp the mask separately, so black parts of an overlay are retained.
    weight = alpha * (mask.astype(np.float32) / 255.0)[:, :, None]
    blended = (1 - weight) * background.astype(np.float32) + weight * warped.ast
    blended = np.clip(blended, 0, 255).astype(np.uint8)
    return blended, warped, mask, H
```

3.3 Load the source and destination images

In [83]:
```
USE_MOUSE = False

building = read_image(image_dir / 'wadham.jpg')
flag = read_image(image_dir / 'flag.png')
flower_poster = read_image(image_dir / 'the_berry_farms_sunflower_field.jpeg')
```

3.4 Set up the two image pairs

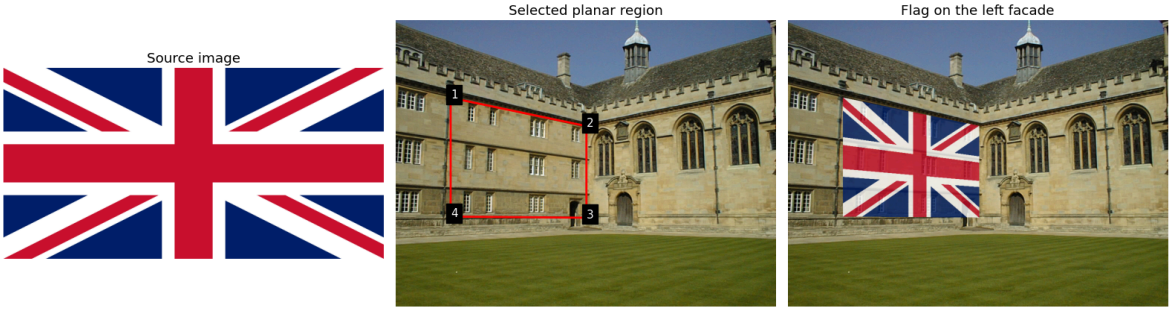
In [84]:
```
overlay_examples = [
    {
        'name': 'Flag on the left facade',
        'image': flag,
        'points': np.float32([[148, 207], [514, 283], [514, 529], [148, 526]]),
        'alpha': 0.80,
        'filename': 'q3_flag_overlay.png'
    },
    {
        'name': 'Sunflower poster on the left wall',
        'image': flower_poster,
        'points': np.float32([[189.87, 210.65], [297.43, 232.59], [285.36, 299.5
        'alpha': 0.85,
        'filename': 'q3_sunflower_overlay.png'
    }
]
```

3.5 Select corners and display each overlay

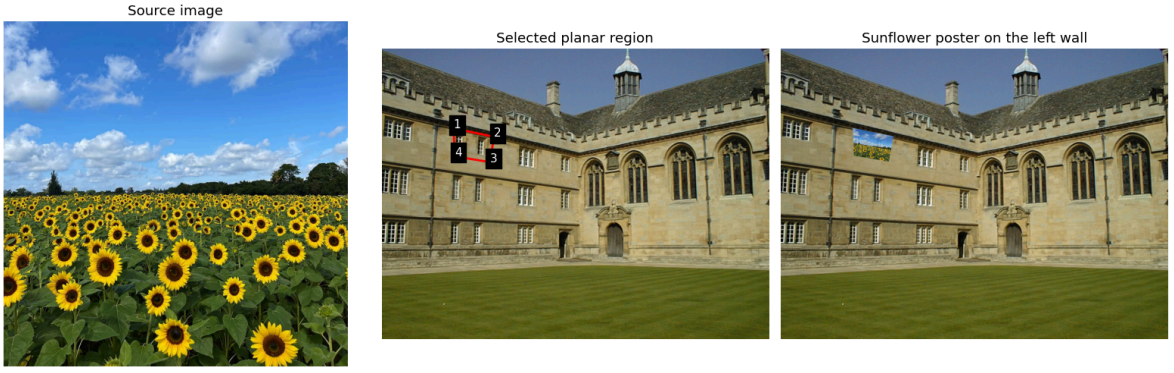
In [85]:
```
overlay_results = []
for example in overlay_examples:
    if USE_MOUSE:
        destination = select_four_points(building, example['name'], example['poi
    else:
        destination = example['points']

    result, warped, mask, H = place_overlay(
        building, example['image'], destination, example['alpha']
    )
    overlay_results.append({'name': example['name'], 'image': result,
                            'points': destination, 'H': H})
    print(example['name'])
    print('Destination points:\n', np.round(destination, 2))
    print('Homography (overlay to building):\n', H)

    fig, axes = plt.subplots(1, 3, figsize=(16, 5))
    show_image(axes[0], example['image'], 'Source image')
    show_image(axes[1], building, 'Selected planar region')
    closed_points = np.vstack((destination, destination[0]))
    axes[1].plot(closed_points[:, 0], closed_points[:, 1], 'r-o', linewidth=2)
```

```
    for i, (x, y) in enumerate(destination):
        axes[1].text(x, y, str(i + 1), color='white', backgroundcolor='black')
    show_image(axes[2], result, example['name'])
    fig.tight_layout()
    save_figure(fig, example['filename'])
    plt.show()
```

```
Flag on the left facade
Destination points:
 [[148. 207.]
 [514. 283.]
 [514. 529.]
 [148. 526.]]
Homography (overlay to building):
 [[ 2.70207637e-01 -3.10405839e-17  1.48000000e+02]
 [ 8.33661672e-02  3.32638165e-01  2.07000000e+02]
 [ 1.54636773e-04 -2.94213628e-19  1.00000000e+00]]
```




```
Sunflower poster on the left wall
Destination points:
 [[189.87 210.65]
 [297.43 232.59]
 [285.36 299.52]
 [193.17 284.16]]
Homography (overlay to building):
 [[8.67962826e-02 2.56983528e-02 1.89869995e+02]
 [2.46697377e-02 8.55138236e-02 2.10649994e+02]
 [4.05135097e-05 1.21163182e-04 1.00000000e+00]]
```



Discussion. The homography changes the source image's perspective so its corners coincide with the chosen quadrilateral. Inside the warped mask the blend is $I_{out}=(1-\alpha)I_{building}+\alpha I_{overlay}$. Outside the mask, the building stays unchanged. An alpha below one lets the wall texture show through. A homography assumes a plane, so recessed windows and parts that protrude from the facade will not have physically exact geometry. The simple blend also does not simulate new shadows or illumination on the poster.

4. PCB alignment and difference images

4(a).1 Read the template

All other PCB images will be aligned to AECC1.jpg.

In [86]:
```
pcb_dir = image_dir / 'pcbs'
template_path = pcb_dir / 'AECC1.jpg'
template = read_image(template_path)
template_gray = cv.cvtColor(template, cv.COLOR_BGR2GRAY)
```

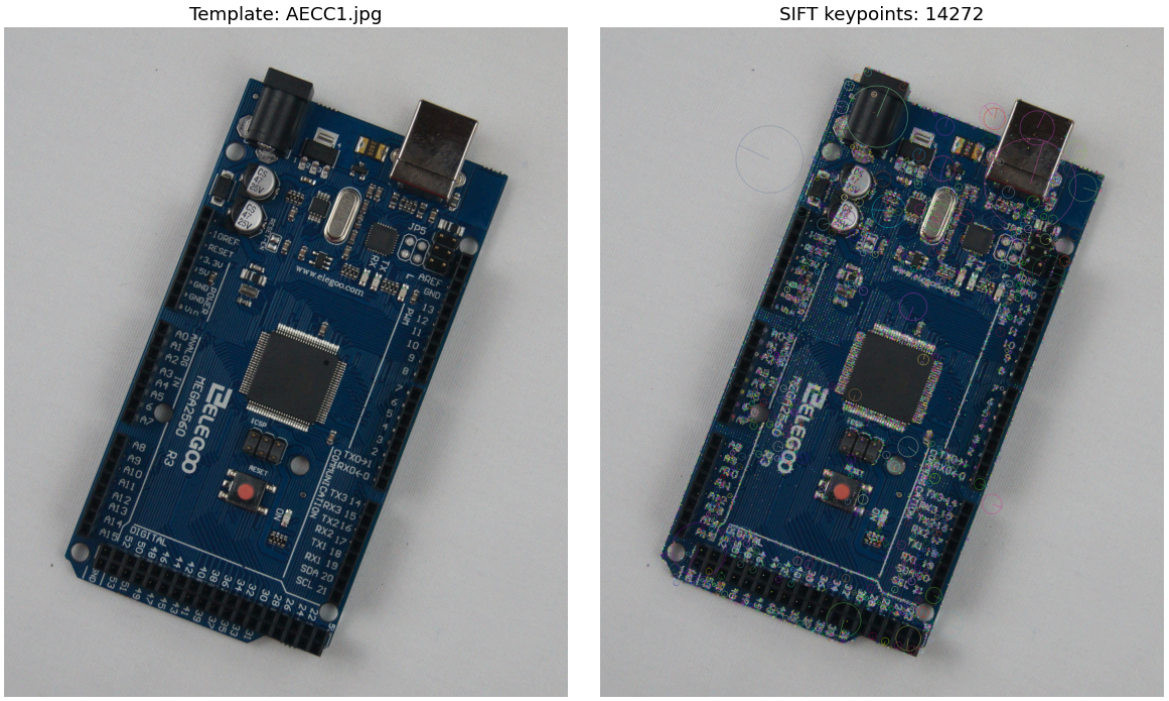
4(a).2 Detect SIFT features in the template

In [87]:
```
sift = cv.SIFT_create()
template_keypoints, template_descriptors = sift.detectAndCompute(template_gray,
if template_descriptors is None or len(template_keypoints) < 4:
    raise ValueError('Not enough SIFT features in the template.')
```

4(a).3 Display the template keypoints

Visualize the locations and scales returned by SIFT.

In [88]:
```
template_features = cv.drawKeypoints(
    template, template_keypoints, None,
    flags=cv.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS
)
fig, axes = plt.subplots(1, 2, figsize=(12, 7))
show_image(axes[0], template, 'Template: AECC1.jpg')
show_image(axes[1], template_features, f'SIFT keypoints: {len(template_keypoints
fig.tight_layout()
save_figure(fig, 'q4_template_sift.png')
plt.show()
```



4(a).4 Define SIFT detection for an input image

This repeats the same detection step for a PCB photograph.

In [89]:
```
def detect_pcb_features(image):
    image_gray = cv.cvtColor(image, cv.COLOR_BGR2GRAY)
    keypoints, descriptors = sift.detectAndCompute(image_gray, None)
    if descriptors is None or len(keypoints) < 4:
        raise ValueError('Not enough SIFT features in this image.')
    return image_gray, keypoints, descriptors
```

4(b).1 Define descriptor matching and the ratio test

Compare each descriptor with the template and reject ambiguous matches.

In [90]:
```
def match_pcb_features(descriptors):
    matcher = cv.BFMatcher(cv.NORM_L2)
    pairs = matcher.knnMatch(descriptors, template_descriptors, k=2)
    good_matches = []
    for pair in pairs:
        if len(pair) == 2:
            first, second = pair
            if first.distance < 0.75 * second.distance:
                good_matches.append(first)
    good_matches.sort(key=lambda match: match.distance)

    if len(good_matches) < 4:
        raise ValueError(f'Only {len(good_matches)} matches passed the ratio tes
    return good_matches
```

4(b).2 Define homography estimation with RANSAC

Convert matched keypoints to coordinate pairs and estimate the mapping to the template.

In [91]:
```
def estimate_pcb_homography(keypoints, good_matches):
    source_points = np.float32([keypoints[m.queryIdx].pt for m in good_matches])
    target_points = np.float32([template_keypoints[m.trainIdx].pt for m in good_
    H, inlier_mask = cv.findHomography(source_points, target_points, cv.RANSAC,
    if H is None or inlier_mask is None or not np.all(np.isfinite(H)):
        raise ValueError('Homography estimation failed.')

    inlier_mask = inlier_mask.ravel().astype(bool)
    if inlier_mask.sum() < 4:
        raise ValueError('Fewer than four homography inliers.')
    return H, inlier_mask, source_points, target_points
```

4(c).1 Define image warping and difference calculation

Warp the board, find the valid overlap and calculate absolute pixel differences.

In [ ]:
```
def warp_pcb_and_difference(image, image_gray, H):
    height, width = template.shape[:2]
    aligned = cv.warpPerspective(image, H, (width, height))
```

```
    source_mask = np.full(image_gray.shape, 255, dtype=np.uint8)
    valid_mask = cv.warpPerspective(source_mask, H, (width, height),flags=cv.INT

    # Exclude a narrow edge where interpolation can mix image and border pixels.
    valid_mask = cv.erode(valid_mask, np.ones((3, 3), dtype=np.uint8))
    valid = valid_mask > 0
    if not valid.any():
        raise ValueError('The warped image has no overlap with the template.')

    difference = cv.absdiff(aligned, template)
    difference[~valid] = 0
    difference_gray = cv.cvtColor(difference, cv.COLOR_BGR2GRAY)
    mean_difference = float(difference[valid].mean())
    return aligned, valid_mask, valid, difference, difference_gray, mean_differe
```

4(b)-(c). Check the fit and draw accepted matches

Measure the inlier reprojection error and draw up to 80 accepted matches.

In [ ]:
```
def check_pcb_alignment(image, keypoints, good_matches, H, inlier_mask, source_p
    projected = cv.perspectiveTransform(source_points[inlier_mask], H)
    reprojection_errors = np.linalg.norm(projected[:, 0, :] - target_points[inli

    # Use every good match for estimation, but draw at most 80 accepted inliers
    inlier_matches = [m for m, keep in zip(good_matches, inlier_mask) if keep]
    match_image = cv.drawMatches(
        image, keypoints, template, template_keypoints, inlier_matches[:80], Non
        matchColor=(0, 255, 0), flags=cv.DrawMatchesFlags_NOT_DRAW_SINGLE_POINTS
    )
    return reprojection_errors, match_image
```

Combine the PCB alignment steps

In [94]:
```
def align_pcb(image):
    image_gray, keypoints, descriptors = detect_pcb_features(image)
    good_matches = match_pcb_features(descriptors)
    H, inlier_mask, source_points, target_points = estimate_pcb_homography(keypo
    aligned, valid_mask, valid, difference, difference_gray, mean_difference = w
        image, image_gray, H
    )
    reprojection_errors, match_image = check_pcb_alignment(
        image, keypoints, good_matches, H, inlier_mask, source_points, target_po
    )

    return {
        'H': H,
        'aligned': aligned,
        'difference': difference,
        'difference_gray': difference_gray,
        'valid_mask': valid_mask,
        'match_image': match_image,
        'keypoint_count': len(keypoints),
        'good_match_count': len(good_matches),
        'inlier_count': int(inlier_mask.sum()),
        'mean_difference': mean_difference,
        'overlap_percent': 100 * valid.mean(),
        'reprojection_rmse': float(np.sqrt(np.mean(reprojection_errors**2)))
    }
```

4(d). Process every supplied PCB image

4(d).1 Find and align all PCB images

Store each successful alignment so the following cells can display it without recomputing SIFT.

In [ ]:
```
pcb_paths = sorted(path for path in pcb_dir.glob('*.jpg') if path.name != templa
if not pcb_paths:
    raise FileNotFoundError('No PCB images were found apart from the template.')

pcb_results = {}
pcb_failures = {}

for path in pcb_paths:
    image = read_image(path)
    try:
        result = align_pcb(image)
    except (ValueError, cv.error) as error:
        pcb_failures[path.name] = str(error)
        print(f'{path.name}: alignment failed - {error}')
        continue

    pcb_results[path.name] = result
    print(f'\n{path.name}: {result["good_match_count"]} ratio-test matches, 'f'{
    print(f'Reprojection RMSE: {result["reprojection_rmse"]:.3f} pixels')
    print(f'Valid overlap: {result["overlap_percent"]:.1f}%')
    print(f'Mean absolute RGB-channel difference: {result["mean_difference"]:.3f
    print('Homography (input to template):\n', result['H'])
```

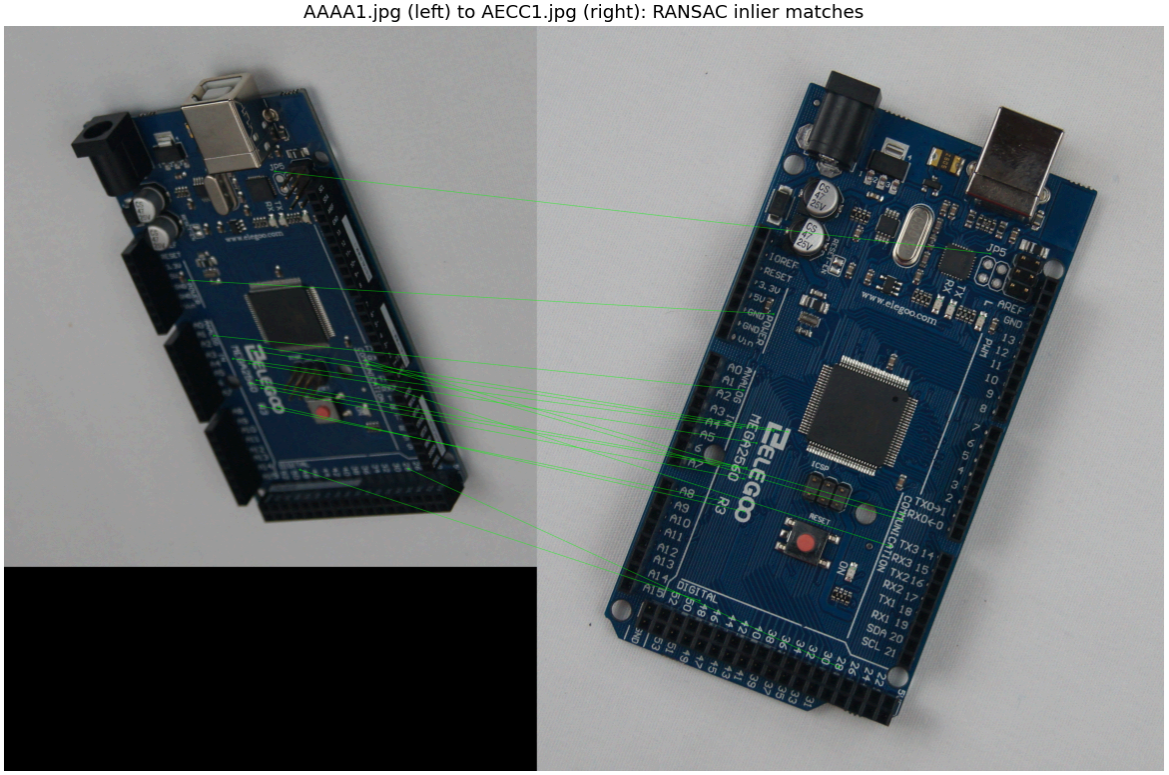
```
AAAA1.jpg: 51 ratio-test matches, 17 RANSAC inliers
Reprojection RMSE: 2.059 pixels
Valid overlap: 88.8%
Mean absolute RGB-channel difference: 29.135 / 255
Homography (input to template):
 [[ 1.18471157e+00 -1.10146759e+00  1.04740107e+03]
 [ 6.30688379e-01  9.47681491e-01 -3.09498515e+02]
 [ 2.29504662e-05 -7.02219946e-05  1.00000000e+00]]

ADAD1.jpg: 369 ratio-test matches, 193 RANSAC inliers
Reprojection RMSE: 1.217 pixels
Valid overlap: 98.0%
Mean absolute RGB-channel difference: 30.690 / 255
Homography (input to template):
 [[ 1.23489975e+00 -1.51865715e-02 -6.25266975e+01]
 [ 1.45361079e-01  1.05663530e+00 -1.54214823e+02]
 [ 5.32794972e-05  5.29653546e-06  1.00000000e+00]]

AECC2.jpg: 4322 ratio-test matches, 3839 RANSAC inliers
Reprojection RMSE: 0.648 pixels
Valid overlap: 98.4%
Mean absolute RGB-channel difference: 3.520 / 255
Homography (input to template):
 [[ 1.00061008e+00 -1.50343341e-02  3.20800275e+01]
 [ 1.65970603e-02  1.00003437e+00 -1.81867146e+01]
 [ 3.53394326e-07  2.57427690e-07  1.00000000e+00]]

AECC3.jpg: 5629 ratio-test matches, 5001 RANSAC inliers
Reprojection RMSE: 0.591 pixels
Valid overlap: 98.9%
Mean absolute RGB-channel difference: 3.295 / 255
Homography (input to template):
 [[ 9.99525742e-01 -1.00206414e-02  2.24982455e+01]
 [ 1.03810349e-02  9.99102775e-01 -1.12692630e+01]
 [-2.02062638e-08  1.33895793e-07  1.00000000e+00]]

AECC4.jpg: 3830 ratio-test matches, 3396 RANSAC inliers
Reprojection RMSE: 0.690 pixels
Valid overlap: 98.5%
Mean absolute RGB-channel difference: 4.066 / 255
Homography (input to template):
 [[ 9.99600986e-01 -1.39817595e-02  3.14182891e+01]
 [ 1.39417406e-02  9.99182659e-01 -1.40739136e+01]
 [ 3.75196875e-08  3.26177790e-07  1.00000000e+00]]
```

4(d).2 Display feature matches for every PCB

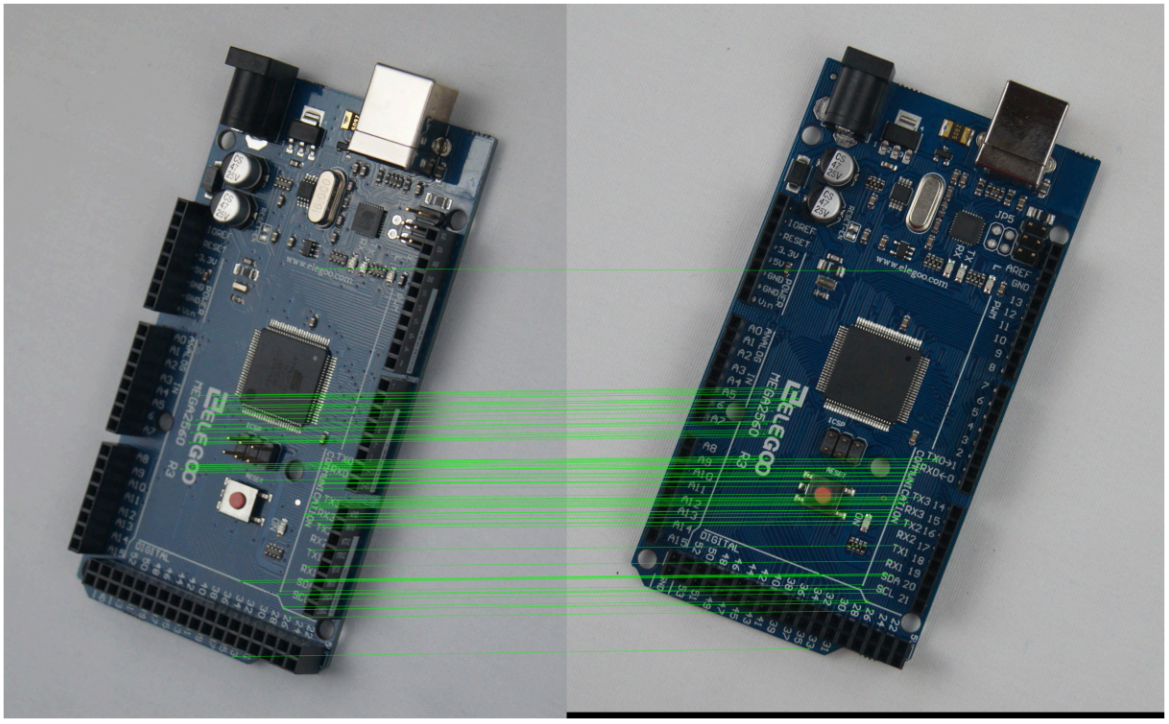
The input image appears on the left and the template on the right.

In [96]:
```
for name, result in pcb_results.items():
    path = pcb_dir / name
    fig, ax = plt.subplots(figsize=(14, 8))
    show_image(ax, result['match_image'], f'{path.name} (left) to AECC1.jpg (rig
    fig.tight_layout()
    save_figure(fig, f'q4_{path.stem}_matches.png')
    plt.show()
```

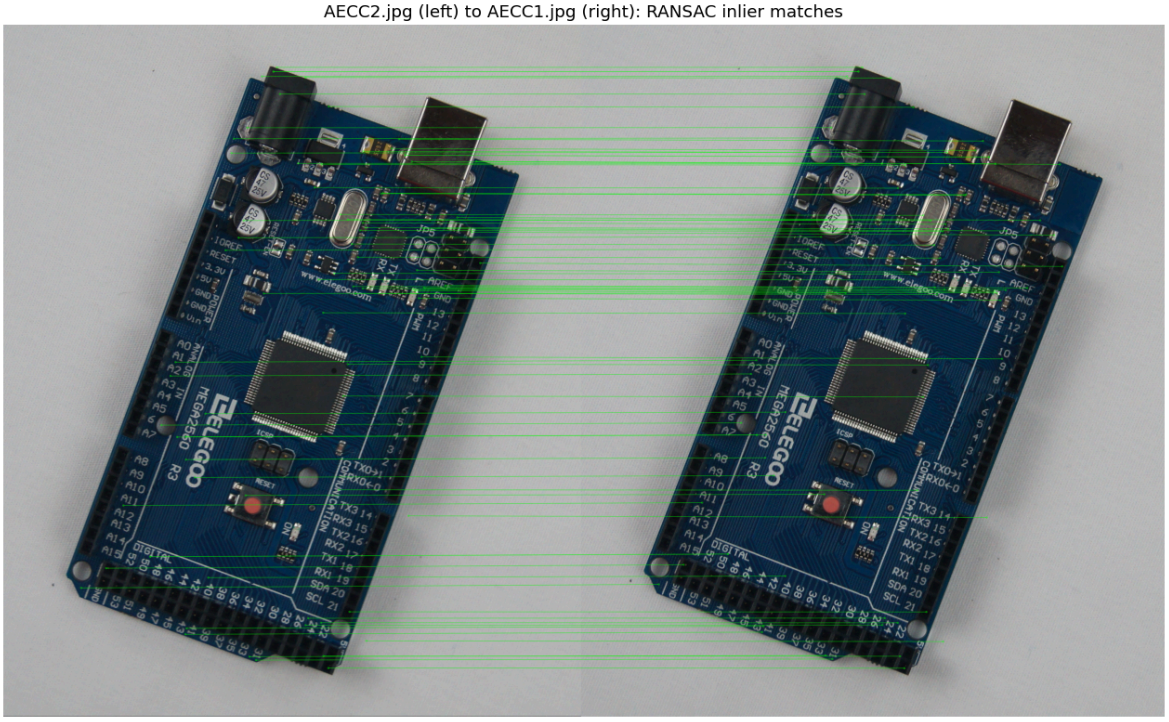
AAAA1.jpg (left) to AECC1.jpg (right): RANSAC inlier matches



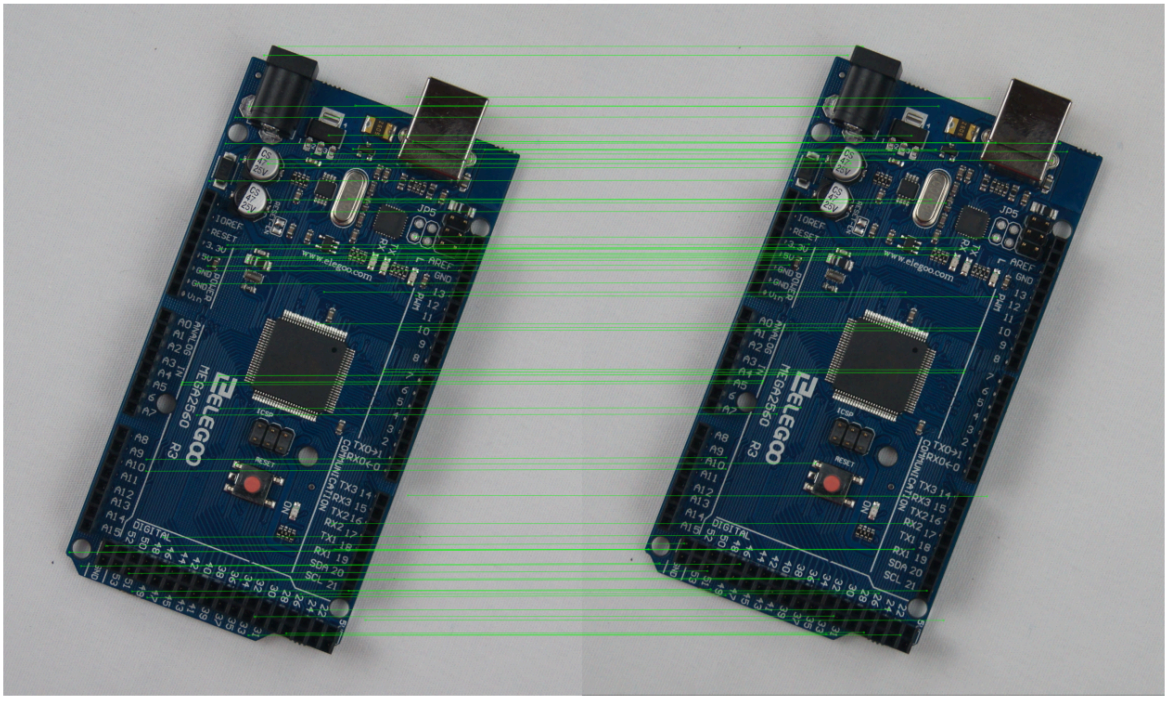
ADAD1.jpg (left) to AECC1.jpg (right): RANSAC inlier matches



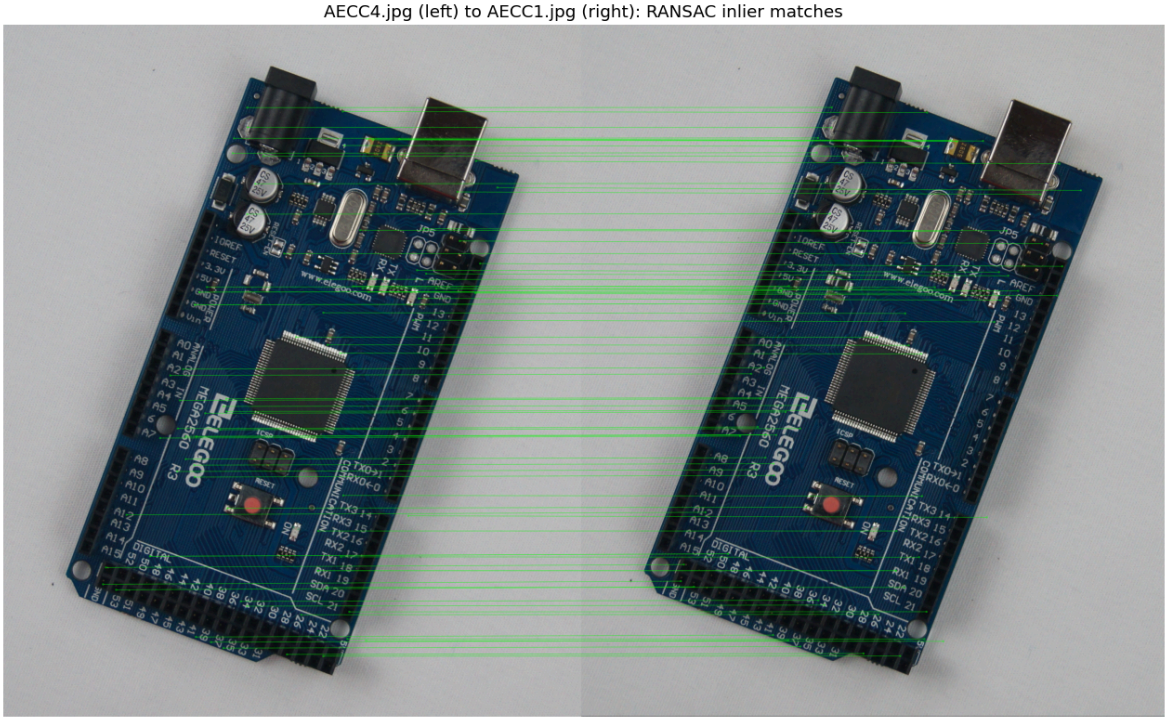
AECC2.jpg (left) to AECC1.jpg (right): RANSAC inlier matches



AECC3.jpg (left) to AECC1.jpg (right): RANSAC inlier matches



AECC4.jpg (left) to AECC1.jpg (right): RANSAC inlier matches

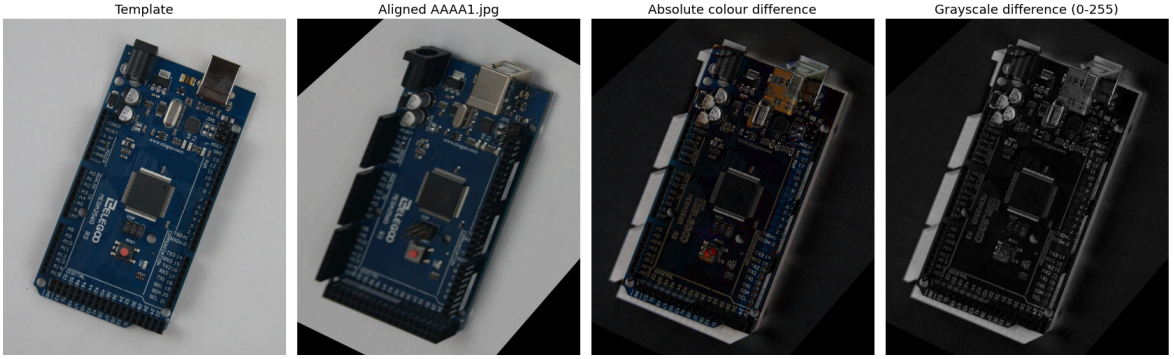


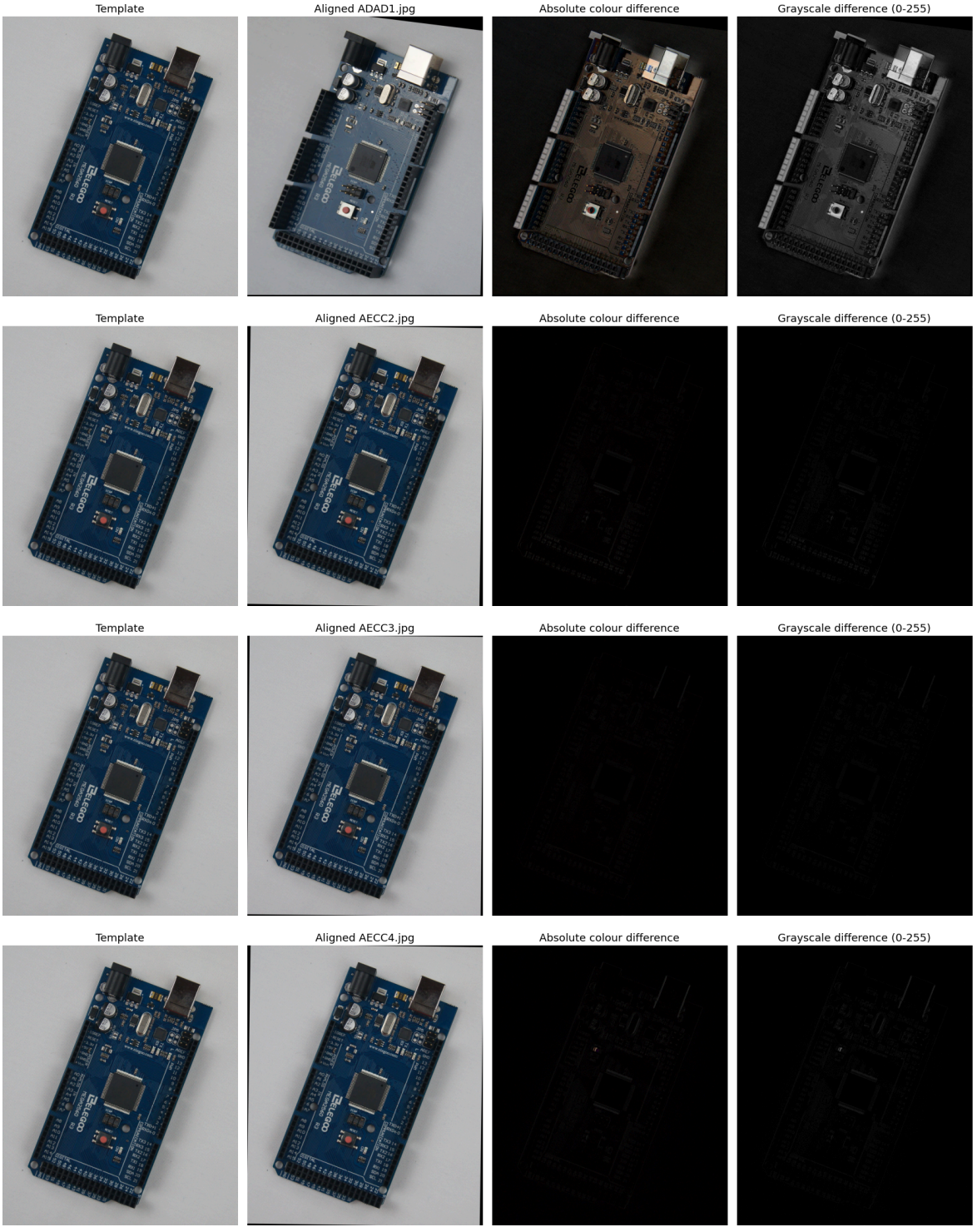
4(d).3 Display aligned boards and their differences

Show each comparison and optionally save the difference image and overlap mask.

In [97]:
```
for name, result in pcb_results.items():
    path = pcb_dir / name
    fig, axes = plt.subplots(1, 4, figsize=(17, 6))
    show_image(axes[0], template, 'Template')
    show_image(axes[1], result['aligned'], f'Aligned {path.name}')
    show_image(axes[2], result['difference'], 'Absolute colour difference')
    show_image(axes[3], result['difference_gray'], 'Grayscale difference (0-255)
    fig.tight_layout()
    save_figure(fig, f'q4_{path.stem}_alignment.png')
    plt.show()

    if SAVE_FIGURES:
        cv.imwrite(str(output_dir / f'q4_{path.stem}_difference.png'), result['d
        cv.imwrite(str(output_dir / f'q4_{path.stem}_valid_mask.png'), result['v
```

Template | Aligned AAAA1.jpg | Absolute colour difference | Grayscale difference (0-255)





4(d).4 Compare alignment statistics

List matches, inliers, geometric error, overlap and average difference.

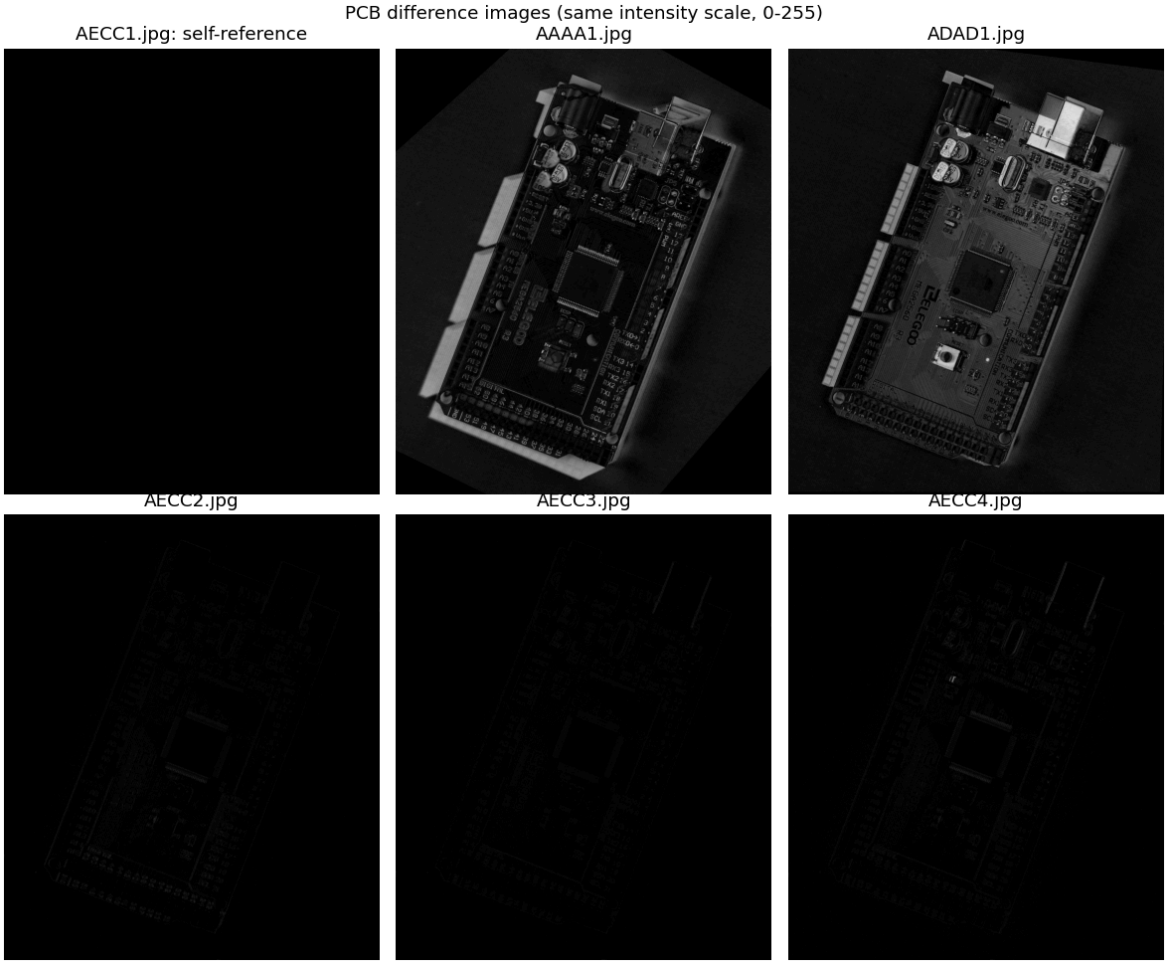
In [98]:
```
print(f'{"Image":<13} {"Matches":>8} {"Inliers":>8} {"RMSE (px)":>11} {"Overlap
for name, result in pcb_results.items():
    print(f'{name:<13} {result["good_match_count"]:>8} {result["inlier_count"]:>
          f'{result["reprojection_rmse"]:>11.3f} {result["overlap_percent"]:>11.
          f'{result["mean_difference"]:>11.3f}')

if pcb_failures:
    print('\nImages still needing a successful alignment:')
    for name, reason in pcb_failures.items():
        print(f'{name}: {reason}')
```

Image	Matches	Inliers	RMSE (px)	Overlap %	Mean diff
AAAA1.jpg	51	17	2.059	88.8	29.135
ADAD1.jpg	369	193	1.217	98.0	30.690
AECC2.jpg	4322	3839	0.648	98.4	3.520
AECC3.jpg	5629	5001	0.591	98.9	3.295
AECC4.jpg	3830	3396	0.690	98.5	4.066

4(d).5 Show all difference images together

Use the same intensity scale for every image.

In [99]:
```
overview_names = [template_path.name] + [path.name for path in pcb_paths]
columns = 3
rows = int(np.ceil(len(overview_names) / columns))
fig, axes = plt.subplots(rows, columns, figsize=(12, 5 * rows), squeeze=False)

for ax, name in zip(axes.flat, overview_names):
    if name == template_path.name:
        show_image(ax, np.zeros_like(template_gray), 'AECC1.jpg: self-reference'
    elif name in pcb_results:
        show_image(ax, pcb_results[name]['difference_gray'], name)
    else:
        ax.text(0.5, 0.5, f'{name}\nAlignment failed', ha='center', va='center')
        ax.axis('off')

for ax in axes.flat[len(overview_names):]:
    ax.axis('off')
fig.suptitle('PCB difference images (same intensity scale, 0-255)')
fig.tight_layout()
save_figure(fig, 'q4_all_difference_images.png')
plt.show()
```



4(e). Why do the USB port, capacitors and reset button show high differences?

The USB port, capacitors and reset button protrude above the PCB, causing parallax that a planar homography cannot fully correct. Changes in reflections and shadows also produce high pixel differences, even when these components are unchanged.